

Project 3: Robust and Privacy-Aware Inference Under Active Attacks: Sharing Useful Features While Protecting Sensitive Attributes

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Project Overview

This project investigates how a system can share useful information with an authorized receiver without revealing sensitive information to an attacker. Each data sample contains a public attribute needed for an allowed task and a private attribute that should remain hidden. The attacker is active: in addition to observing transmissions and trying to infer the private attribute, it may disrupt, block, or manipulate selected messages. Students will build a simulation-based environment and design data-sharing strategies that preserve the authorized task under attack while limiting sensitive information leakage.

Project Goals

- **Public and Private Tasks:** Define a data source with two related attributes, such as an activity needed for a service and an identity or location that should remain private.
- **Privacy-Aware Data Sharing:** Implement methods that release selected samples, features, or transformed representations so the authorized receiver can perform the public task while the private attribute is difficult to infer.
- **Active Attack Simulation:** Develop attacker models that combine private-attribute inference with actions such as message dropping, jamming, replay, or data manipulation.
- **Robust Strategy Evaluation:** Compare defensive strategies across different attacker strengths, communication conditions, and relationships between public and private attributes.

The project aims to demonstrate how privacy and robustness can be designed together rather than treating information leakage and active attacks as separate problems.

Motivation and Significance

Modern systems often process data that supports a useful service while also containing sensitive information. For example, wearable data may support activity recognition while revealing identity, and smart-building data may support energy control while revealing occupancy patterns. Protecting these systems becomes harder when an attacker can both learn from the communication and interfere with it. This creates several important challenges:

- **Task Utility Versus Attribute Privacy:** Features that are useful for the authorized task may also reveal a sensitive attribute because the two are statistically related.
- **Passive and Active Threats:** A defense that hides information from a passive observer may fail when the attacker can selectively disrupt or modify the transmitted data.
- **Adaptive System Design:** The best data-sharing strategy may depend on the observed attack, the reliability of the channel, and the accuracy required by the authorized application.

Research Challenge: The system must balance three objectives at once: accurate public-task inference for the authorized receiver, poor private-attribute inference for the attacker, and continued operation under active interference. Students will investigate whether feature selection, randomization, redundancy, attack detection, or adaptive transmission can provide a useful operating region across these competing goals.

Learning Outcomes

Students will:

- Gain hands-on experience in security-oriented machine learning and adversarial system simulation.
- Implement and evaluate classifiers for both an allowed task and a sensitive-attribute inference attack.
- Model active attacks and study their effects on privacy, reliability, and task performance.
- Develop experimental skills through reproducible software, parameter studies, and clear visualization of trade-offs.

Background Required

- Programming experience.
- Basic knowledge of probability, statistics, and data analysis.
- Prior knowledge of wireless communications is not required nor expected.

Expected Outcomes

- A simulator containing a data source, privacy-aware sender, authorized receiver, and active attacker.
- An experimental comparison of baseline and defensive strategies using public-task accuracy, private-attribute leakage, attack success, and communication cost.
- A written report and presentation describing the system design, threat model, results, limitations, and promising directions for further research or a master's thesis.